T&Cs Analyzer

(Analyzing T&Cs using AI for Risk Analysis)

STAT 541: Final Project Proposal

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Question:

Have you ever read the T&Cs before using a product?

Not so Fun Facts:

1. Most of the software products we buy are just licenses.

2. Even if you have paid for a product license, the vendor has full authority to revoke it. For example: Ubisoft

3. Some corporations have huge penalties for ending subscription services early. For example: Adobe

Problem Statement

Users agree to contracts they do not understand.

Terms & Conditions are:

- Extremely long
- Hard to understand (legal jargon)
- Rarely read by users

Users often:

- Accept without knowing risks

Motivation

Real-world Impact:

- Privacy risks (data sharing)
- Hidden charges / subscriptions
- Legal liabilities

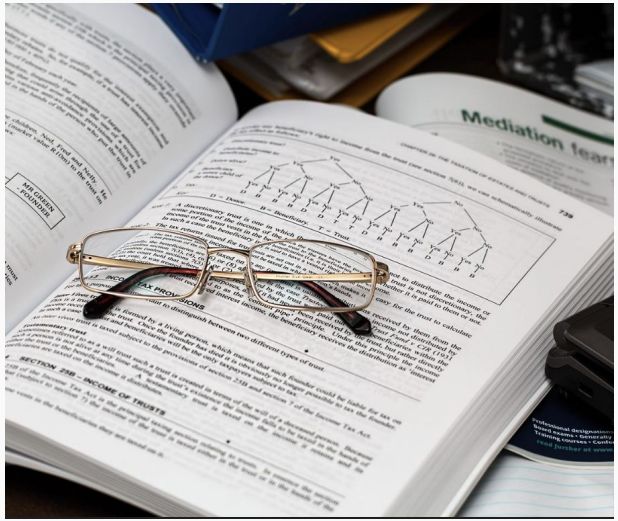
Growing Importance:

- Digital platforms everywhere

Literature

1. **RELexED: Retrieval-Enhanced Legal Summarization:**
   It is a study in 2025 that shows retrieval-enhanced models improve legal summarization quality.

2. **RAG for legal systems:**
   In this paper, the key takeaway is that the retrieval quality is critical for legal AI reliability.



Limitations of Existing Solutions

Current Summarizers:

- Only shorten text

- Do NOT:
  - Highlight risky clauses
  - Categorize information
  - Provide actionable insights

The Solution + Project Objective

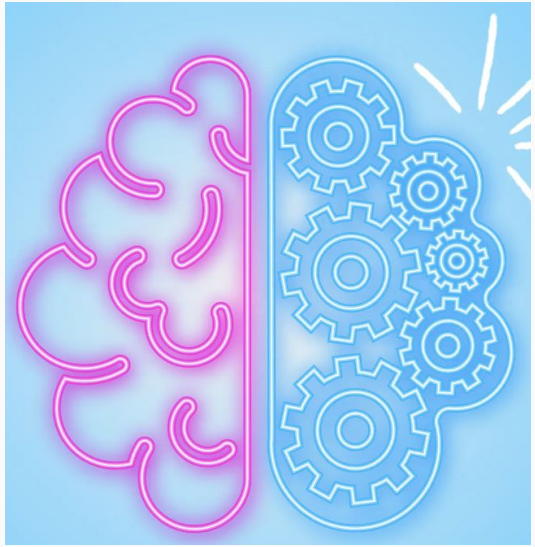
Build an AI agent that:

- Summarizes T&Cs in plain English
- Identifies key clauses:
  - Privacy
  - Payments
  - Termination
- Flags risky content
- (Optional) Gives recommendations

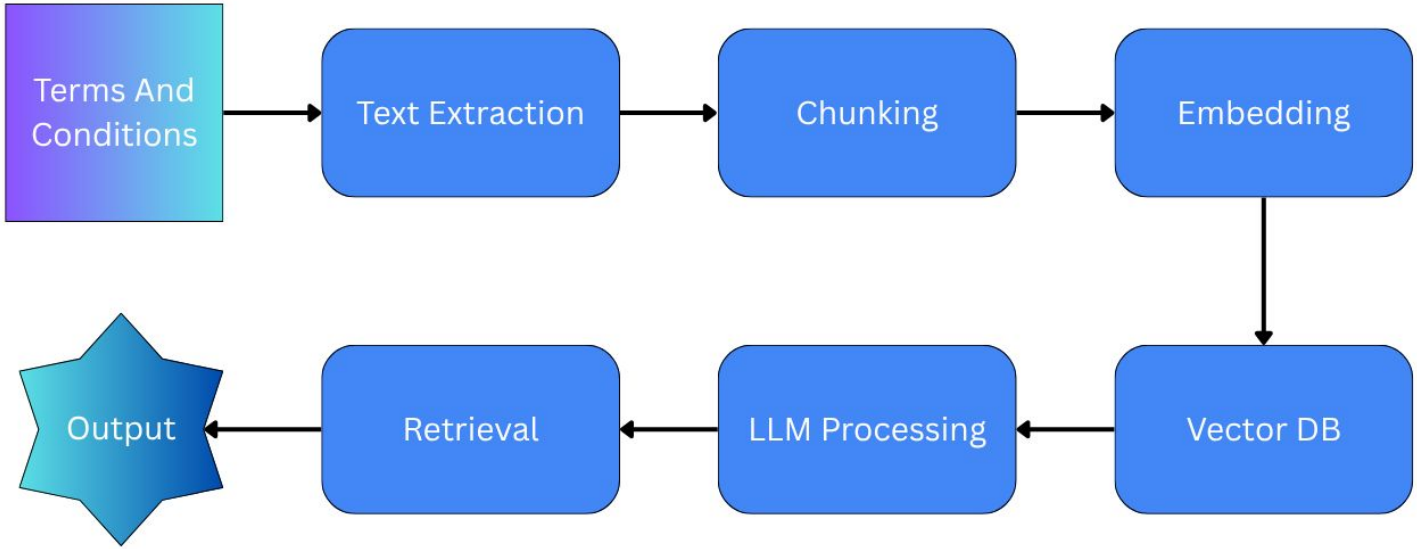
Methodology + System Design

Proposed Solution Overview

- AI-powered document analysis system
- Uses:
  - LLM + Retrieval (RAG)
- Output:
  - Summary + Risk Analysis



System Architecture



Key Components

Break into modules:

- Document Loader (PDF/Text)
- Chunking & Preprocessing
- Embedding Model
- Vector Database (FAISS)
- LLM (OpenAI API)
- Output Formatter

Output format:

Summary

+

Risk Flags

AI Agent Functionality

Agent tasks:

- Summarization
- Clause Extraction
- Risk Classification

Optional:

- Multi-step reasoning (ReAct-style)

  “Find payment terms → analyze → summarize”

Progressive Disclosure: (Complex to Simple)

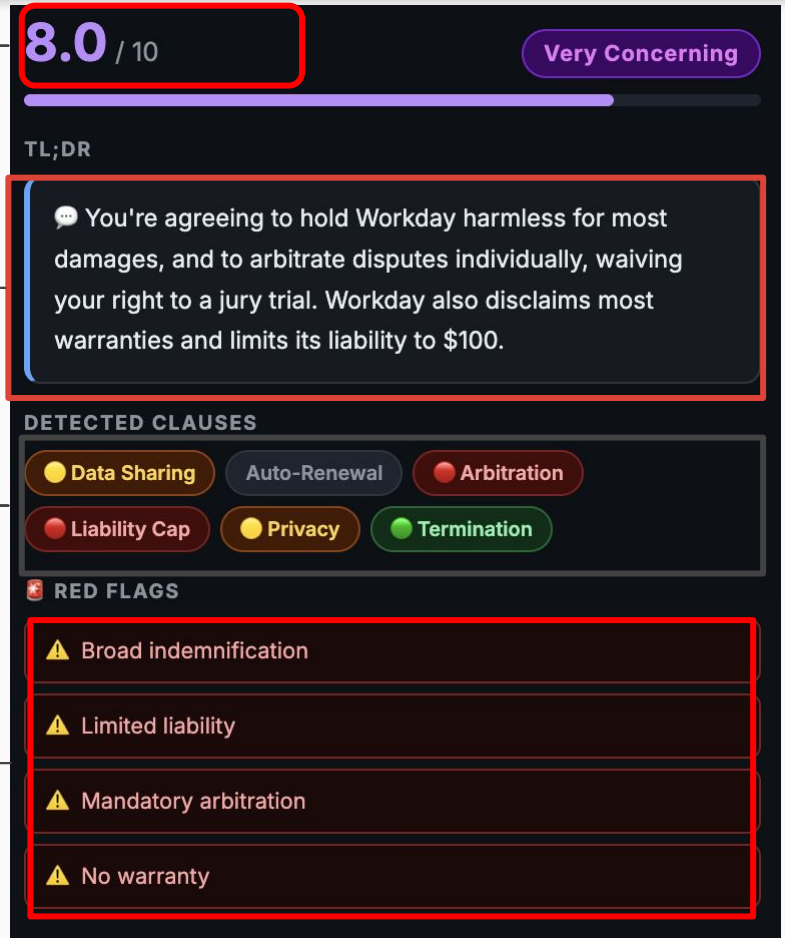
Workday T&Cs:
The following output is the analysis of the T&Cs using our model with layered approach.

Layer - 1

Layer - 2

Layer - 3

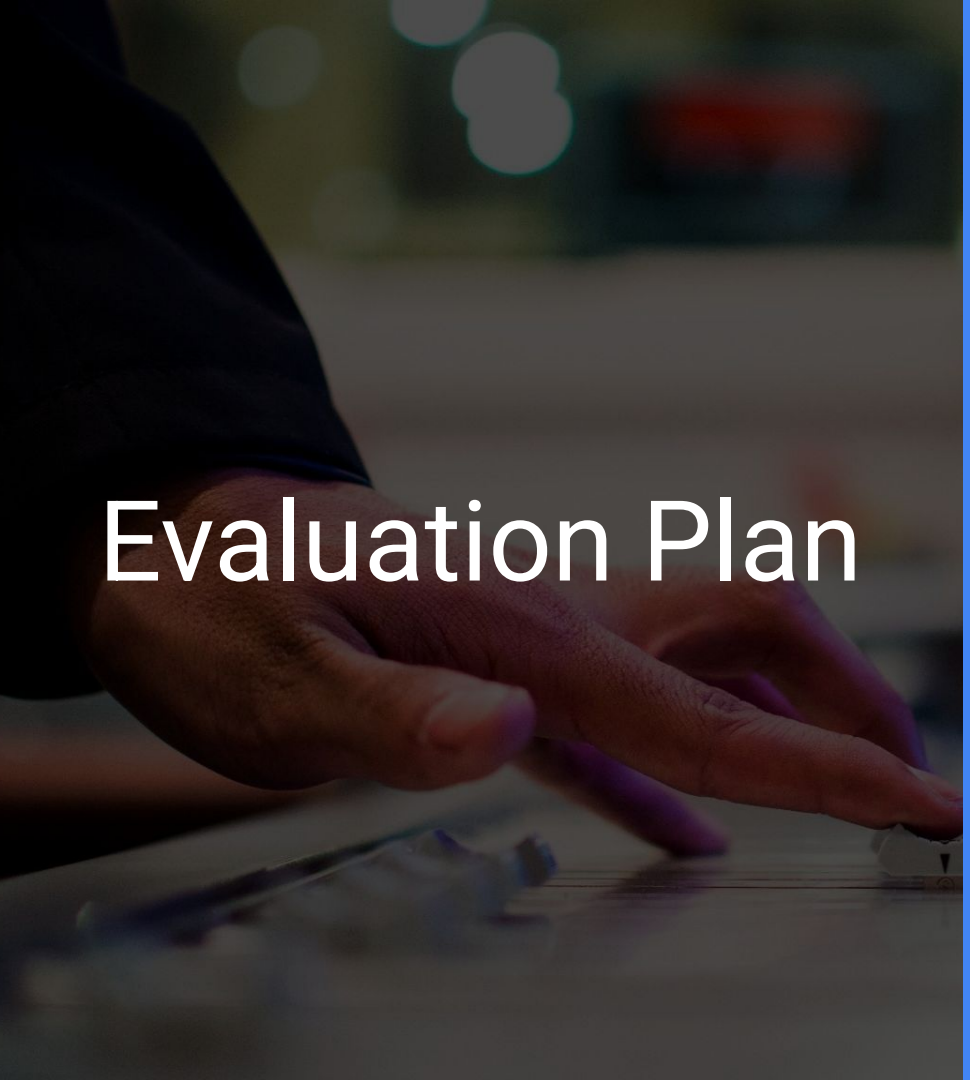
Layer - 4



Tools and Technologies

Component	Technology	Justification
LLM Framework	LangChain 0.2+	Industry-standard, strong RAG primitives
Vector Database	FAISS	Zero-dependency, local, fast for doc-scale indices
Embeddings	OpenAI text-embedding-3-small	High quality, low cost (~$0.00002/1K tokens)
LLM (primary)	Groq llama-3.3-70b-versatile	Free tier, fast inference, reliable JSON output
LLM (alternative)	OpenAI GPT-4o-mini	Strong structured output, low cost
PDF Processing	PyPDF	Lightweight, no system dependencies
Web UI	Streamlit	Rapid prototyping, suitable for academic demos
Browser Extension	Vanilla JS + Manifest V3	No build toolchain, easy to load unpacked

Evaluation, Contributions and Plans



Evaluation Plan

We evaluate:

1. Summary quality
2. Risk detection accuracy
3. Usefulness to users

Evaluation Methods

Compare with baseline:

- Simple summarization vs our system

Human evaluation:

- Clarity
- Correctness

Metrics:

- ROUGE (optional)
- Coverage of key clauses

Anticipated Contributions:

Name	Contributions
Vaibhav Sharma	Using AI agent for legal text (T&Cs) understanding
Aryan Pandey	Structured risk-aware summaries
Khawaja Abbas	Implementation of RAG pipeline and LLM-based reasoning

Limitations and Future Work

Limitations:

1. LLM hallucinations
2. Legal interpretation not perfect
3. Depends on quality of input

Future Work:

1. Multi-agent system
2. Better legal accuracy
3. Personalized recommendations

Thank You And Enjoy The Demonstration